

Git Commands & Concepts Summary

Here's everything we covered in this chat session, summarized for easy reference:

◆ 1. Installing & Configuring Git

- Check if Git is installed:

```
sh

git --version
```



- Configure Git with your GitHub username & email:

```
sh

git config --global user.name "your_username" git config --global user.email "your_email@example.com"
```



- Verify your Git configuration:

```
sh

git config --list
```



◆ 2. Cloning a GitHub Repository

To download a project from GitHub to your local system:

```
sh
```



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```
git clone https://github.com/your_username/repository_name.git
```

Then move into the folder:

```
sh
```



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```
cd repository_name
```

◆ 3. Working with Branches

Creating a New Branch

- Create a branch but don't switch to it:

```
sh
```



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```
git branch branch_name
```

- Create & switch to a new branch in one step:

```
sh
```



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```
git checkout -b branch_name
```

or (newer Git versions):

```
sh
```

```
git switch -c branch_name
```



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Listing Branches

- Show local branches:

```
sh
```

```
git branch
```



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- Show local & remote branches:

```
sh
```

```
git branch -a
```



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- Show only remote branches:

```
sh
```

```
git branch -r
```



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Switching Between Branches

```
sh
```

```
git checkout branch_name
```



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or

```
sh
```



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```
git switch branch_name
```

Renaming a Branch

```
sh
```

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```
git branch -M new_branch_name
```

◆ 4. Pushing and Pulling Changes

Pushing Code to GitHub

- First-time push for a new branch:

```
sh
```

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```
git push --set-upstream origin branch_name
```

or

```
sh
```

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```
git push -u origin branch_name
```

- After setting upstream, simply use:

```
sh
```

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```
git push
```

Pulling Updates from GitHub

```
sh
```

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```
git pull origin branch_name
```

or just

```
sh
```

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```
git pull
```

if upstream is already set.

◆ 5. Merging Branches

Merging main into Jason

```
sh
```

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```
git checkout Jason # Switch to the branch you want to update git merge main # Merge main into Jason
```

If there are conflicts, manually resolve them, then:

```
sh
```

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```
git add . git commit -m "Resolved merge conflicts"
```

◆ 6. Undoing a Merge

Undo Merge Before Pushing

```
sh
```

```
git reset --hard HEAD~1
```



Undo Merge After Pushing (Force Push Required!)

```
sh
```

```
git reset --hard HEAD~1 git push --force
```



Undo Merge But Keep Changes

```
sh
```

```
git reset --soft HEAD~1
```



◆ 7. Understanding origin in Git

- origin is the default name for the remote repository (GitHub, GitLab, etc.).
- Check your remotes:

```
sh
```



```
git remote -v
```

- Change the remote URL:

```
sh
```



```
git remote set-url origin new_repository_url.git
```

🚀 Key Takeaways

- ✓ Git helps manage versions of your project.
- ✓ origin is your default GitHub repository link.
- ✓ Always checkout or switch to the correct branch before making changes.
- ✓ Use `git push --set-upstream origin branch_name` once, then just `git push`.
- ✓ Use `git merge` to combine branches.
- ✓ If you mess up a merge, you can `git reset --hard HEAD~1` to undo.